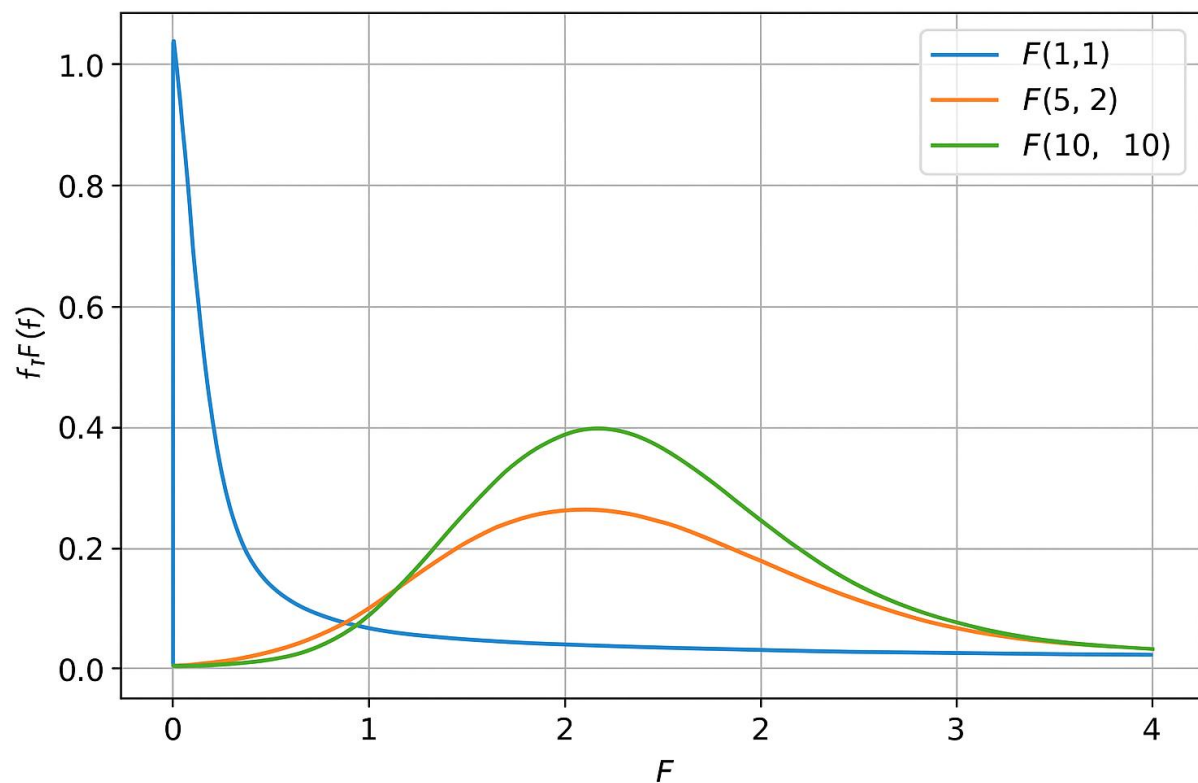


QUESTION

Derive the probability density function for F-distribution.

Probability density function of F-distribution is defined as a ratio of two chi-squared random variable, each divided by its degrees of freedom.

The probability density function (PDF) of an F-distribution; describes the likelihood of different values occurring for a random variable following this distribution.



Theorem: Let F be a random variable following an F-distribution;

$$F \sim F(u, v). \dots\dots\dots (1)$$

Then, the probability density function of F is

$$f_F(f) = \frac{\Gamma\left(\frac{u+v}{2}\right)}{\Gamma\left(\frac{u}{2}\right) \cdot \Gamma\left(\frac{v}{2}\right)} \cdot \left(\frac{u}{v}\right)^{\frac{v}{2}} \cdot f^{\frac{u}{2}-1} \cdot \left(\frac{u}{v}f + 1\right)^{-\frac{u+v}{2}} \dots\dots\dots (2)$$

Proof: An F- distributed random variable is defined as the ratio of two chi-squared random variable, divided by their degree of freedom

$$X \sim \chi^2(u), Y \sim \chi^2(v) \Rightarrow F = \frac{X/u}{Y/v} \sim F(u, v) \dots\dots\dots(3)$$

Where X and Y are independent of each other.

The probability density function of the chi-squared distribution is

$$f_X(x) = \frac{1}{\Gamma\left(\frac{u}{2}\right) \cdot 2^{\frac{u}{2}}} \cdot x^{\frac{u}{2}-1} \cdot e^{-\frac{x}{2}} \dots\dots\dots (4)$$

Define the random variable F and Was function of X and Y

$$F = \frac{X}{Y} FW$$

$$W = Y, \dots\dots\dots (5)$$

Such that the inverse function X and Y in terms of F and W are

$$X = \frac{u}{v} FW$$

$$Y = W \dots\dots\dots (6)$$

This implies the following Jacobian matrix and determinant;

$$J = \begin{bmatrix} \frac{dX}{dF} & \frac{dX}{dW} \\ \frac{dY}{dF} & \frac{dY}{dW} \end{bmatrix} = \begin{bmatrix} \frac{u}{v} W & \frac{u}{v} F \\ 0 & 1 \end{bmatrix} \dots\dots\dots (7)$$

$$|J| = \frac{u}{v} W.$$

Because X and Y are independent, the joint density of X and Y is equal to the product of the marginal densities;

$$f_{X,Y}(x,y) = f_X(x) \cdot f_Y(y) \dots\dots\dots (8)$$

With the probability density function of an invertible function, the joint density of F and W can be derived as;

$$f_{F,W}(f,w) = f_{X,Y}(x,y) \cdot |J| \dots\dots\dots (9)$$

Substituting (6) into (4), and then with (7)

Into (9), we get;

$$\begin{aligned} f_{F,W}(f,w) &= f_X\left(\frac{u}{v}fw\right) \cdot f_Y(w) \cdot |J| \\ &= \frac{1}{\Gamma\left(\frac{u}{2}\right) \cdot 2^{u/2}} \cdot \left(\frac{u}{v}fw\right)^{\frac{u}{2}-1} \cdot e^{-\frac{1}{2}\left(\frac{u}{v}fw\right)} \cdot \frac{1}{\Gamma\left(\frac{v}{2}\right) \cdot 2^{v/2}} \cdot w^{\frac{v}{2}-1} \cdot e^{-\frac{w}{2}} \cdot \frac{u}{v}w \\ &= \frac{\left(\frac{u}{v}\right)^{\frac{u}{2}} \cdot f^{\frac{u}{2}-1}}{\Gamma\left(\frac{u}{2}\right) \cdot \Gamma\left(\frac{v}{2}\right) \cdot 2^{(u+v)/2}} \cdot w^{\frac{u+v}{2}-1} \cdot e^{-\frac{w}{2}\left(\frac{u}{v}f+1\right)} \end{aligned} \quad (10)$$

The marginal density of F can now be obtained by integrating out W:

$$\begin{aligned} f_F(f) &= \int_0^\infty f_{F,W}(f,w) dw \\ &= \frac{\left(\frac{u}{v}\right)^{\frac{u}{2}} \cdot f^{\frac{u}{2}-1}}{\Gamma\left(\frac{u}{2}\right) \cdot \Gamma\left(\frac{v}{2}\right) \cdot 2^{(u+v)/2}} \cdot \int_0^\infty w^{\frac{u+v}{2}-1} \cdot \exp\left[-\frac{1}{2}\left(\frac{u}{v}f+1\right)w\right] dw \\ &= \frac{\left(\frac{u}{v}\right)^{\frac{u}{2}} \cdot f^{\frac{u}{2}-1}}{\Gamma\left(\frac{u}{2}\right) \cdot \Gamma\left(\frac{v}{2}\right) \cdot 2^{(u+v)/2}} \cdot \frac{\Gamma\left(\frac{u+v}{2}\right)}{\left[\frac{1}{2}\left(\frac{u}{v}f+1\right)\right]^{(u+v)/2}} \cdot \int_0^\infty \frac{\left[\frac{1}{2}\left(\frac{u}{v}f+1\right)\right]^{(u+v)/2}}{\Gamma\left(\frac{u+v}{2}\right)} \cdot w^{\frac{u+v}{2}-1} \cdot \exp\left[-\frac{1}{2}\left(\frac{u}{v}f+1\right)w\right] dw \end{aligned} \quad (11)$$

At this point, we can recognize that the integrated is equal to the probability density function of a gamma distribution with

$$a = \frac{u+v}{2} \quad \text{and} \quad b = \frac{1}{2} \left(\frac{u}{v} f + 1 \right) , \quad (12)$$

And because a probability density function integrates to one, we finally have:

$$\begin{aligned} f_F(f) &= \frac{\left(\frac{u}{v}\right)^{\frac{u}{2}} \cdot f^{\frac{u}{2}-1}}{\Gamma\left(\frac{u}{2}\right) \cdot \Gamma\left(\frac{v}{2}\right) \cdot 2^{(u+v)/2}} \cdot \frac{\Gamma\left(\frac{u+v}{2}\right)}{\left[\frac{1}{2}\left(\frac{u}{v}f + 1\right)\right]^{(u+v)/2}} \\ &= \frac{\Gamma\left(\frac{u+v}{2}\right)}{\Gamma\left(\frac{u}{2}\right) \cdot \Gamma\left(\frac{v}{2}\right)} \cdot \left(\frac{u}{v}\right)^{\frac{u}{2}} \cdot f^{\frac{u}{2}-1} \cdot \left(\frac{u}{v}f + 1\right)^{-\frac{u+v}{2}} . \end{aligned} \quad (13)$$

Refference:

1 .”Probability and statistics for Engineers and scientists”

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2. statisticsmatt (2018): “Statistical Distributions: